Experiment -8

Python Program To Implement Depth -First Search

Code:

```
def dfs(graph, node, visited=None):
    if visited is None:
        visited = set()

    if node not in visited:
        print(node, end=" ")
        visited.add(node)

        for neighbor in graph[node]:
            dfs(graph, neighbor, visited)

graph = {
    'A': ['B', 'C'],
    'B': ['D', 'E'],
    'C': ['F'],
    'D': [],
    'E': [],
    'F': []
}

print("DFS Traversal:")
dfs(graph, 'A')
```

```python
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    if visited is None:
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    if node not in visited:
        print(node, end=" ")
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        for neighbor in graph[node]:
            dfs(graph, neighbor, visited)

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print("DFS Traversal:")
dfs(graph, 'A')
```

```
DFS Traversal:
A B D E C F

...Program finished with exit code 0
Press ENTER to exit console.
```